Introduction to Climate Modeling

Week 6

Dhruv Balwada · Columbia Climate School · Fall 2026

Energy input and heat storage

Last week: forcing and feedback

$$N' = F + \alpha T'$$

Net radiation N' supplies additional energy to the climate system.

One-box heat storage

$$E' = CT' \qquad N' = dE'/dt = C\, dT'/dt$$

Assume one reservoir with constant heat capacity C per unit Earth area.

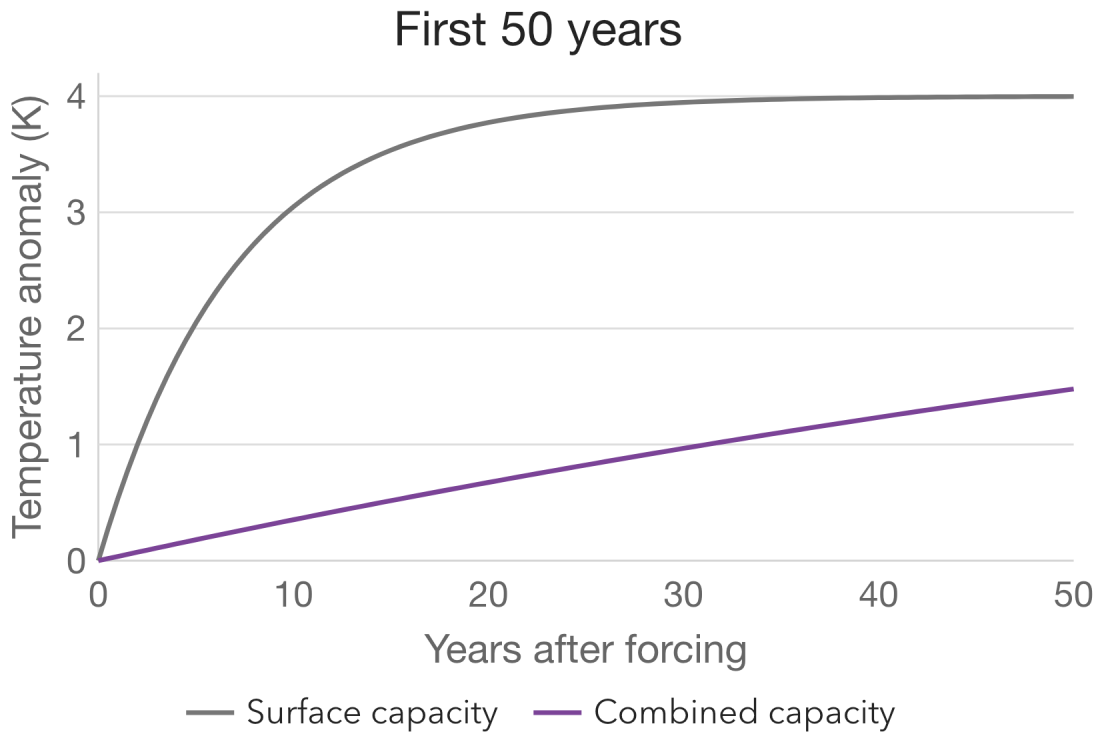
Energy conservation gives the temperature equation

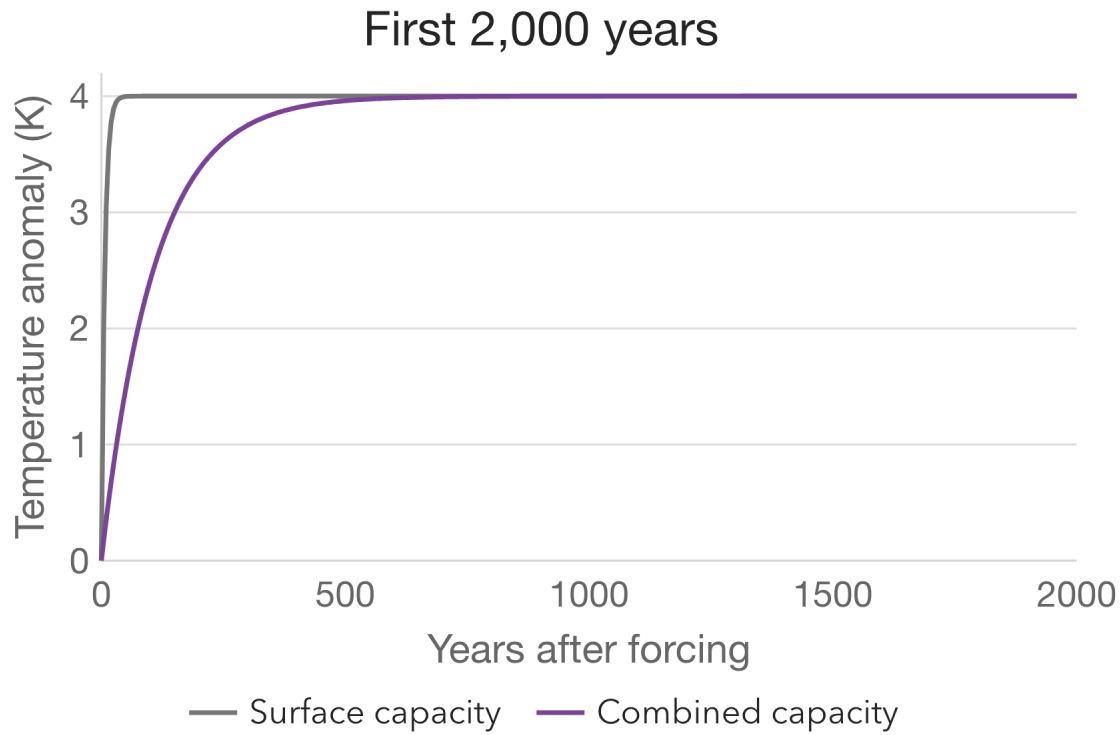
$$C\, dT'/dt = F + \alpha T'$$

N': W m^{-2} E': J m^{-2} C: J m^{-2} K^{-1}
T' is warming from the reference climate. A restoring feedback has $\alpha < 0$.

One-box models with different heat capacities

One reservoir: $C\,dT'/dt = F + \alpha T'$ $\quad \tau = -C/\alpha$





Which model warms faster? Do they reach the same final temperature?

Same $F = 4$ W m^{-2} and $\alpha = -1$ W m^{-2} K^{-1}. Adjustment times: about 7 and 108 years.
Heat capacities: surface 2.2×10^8, combined 3.42×10^9 J m^{-2} K^{-1}, per total Earth area.

Why separate surface and deep ocean?

Observations: Heat accumulates in the upper ocean and at depth. This motivates tracking interior heat storage.

Physical picture: The surface has a smaller heat capacity. The larger ocean interior exchanges heat with it gradually.

Modeling question: Can coupling these reservoirs produce rapid initial surface warming followed by slow adjustment?

The model will show how heat capacity and exchange together set the adjustment timescales.

Sources: NOAA Climate.gov (2025). Held et al. (2010). Geoffroy et al. (2013).

Two-box ocean

Fast surface response, slow ocean adjustment

- **Surface reservoir:** the upper ocean and rapidly coupled atmosphere and land, represented by T_s' and heat capacity C_s.
- **Deep reservoir:** the ocean interior, represented by T_d' and a larger heat capacity C_d.
- **Exchange:** as the surface warms, some incoming energy enters the interior and delays surface warming.

Where does the energy go while the planet is still gaining heat?

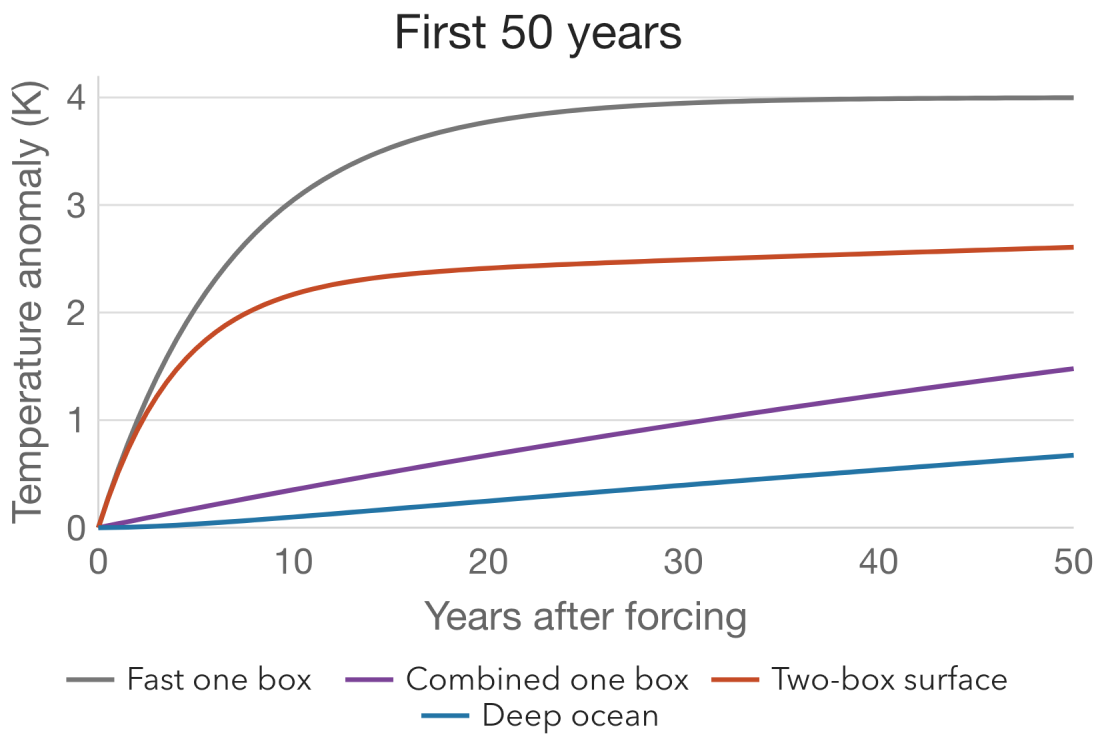
Two-box ocean: coupled energy budgets

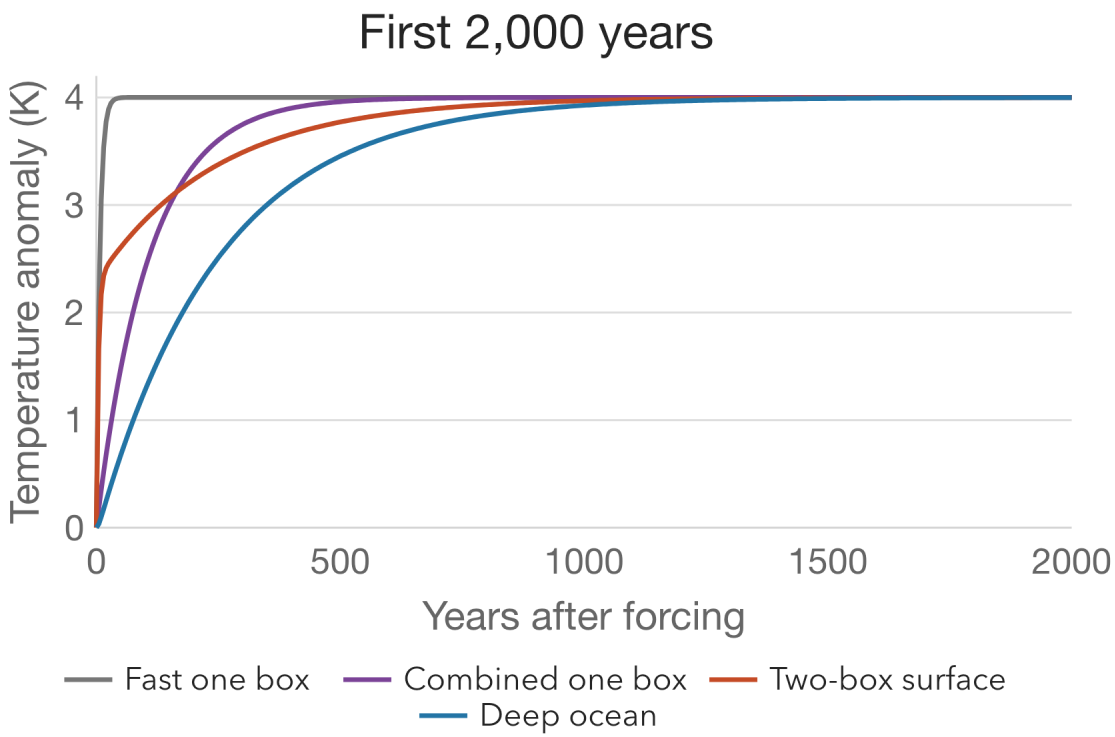
$$C_s \, dT_s'/dt = F + \alpha T_s' - Q'$$
$$C_d \, dT_d'/dt = Q'$$
$$Q' = \gamma(T_s' - T_d')$$

- **Q':** additional heat transfer from the surface to the deep box, in W m^{-2}.
- **$\gamma \geq 0$:** exchange coefficient, in W m^{-2} K^{-1}. A larger temperature-anomaly difference drives more transfer.
- **C_s and C_d:** heat capacities in J m^{-2} K^{-1}, both per total Earth area.

The surface loses exactly the energy that the deep box gains.

Coupled reservoirs: fast and slow warming

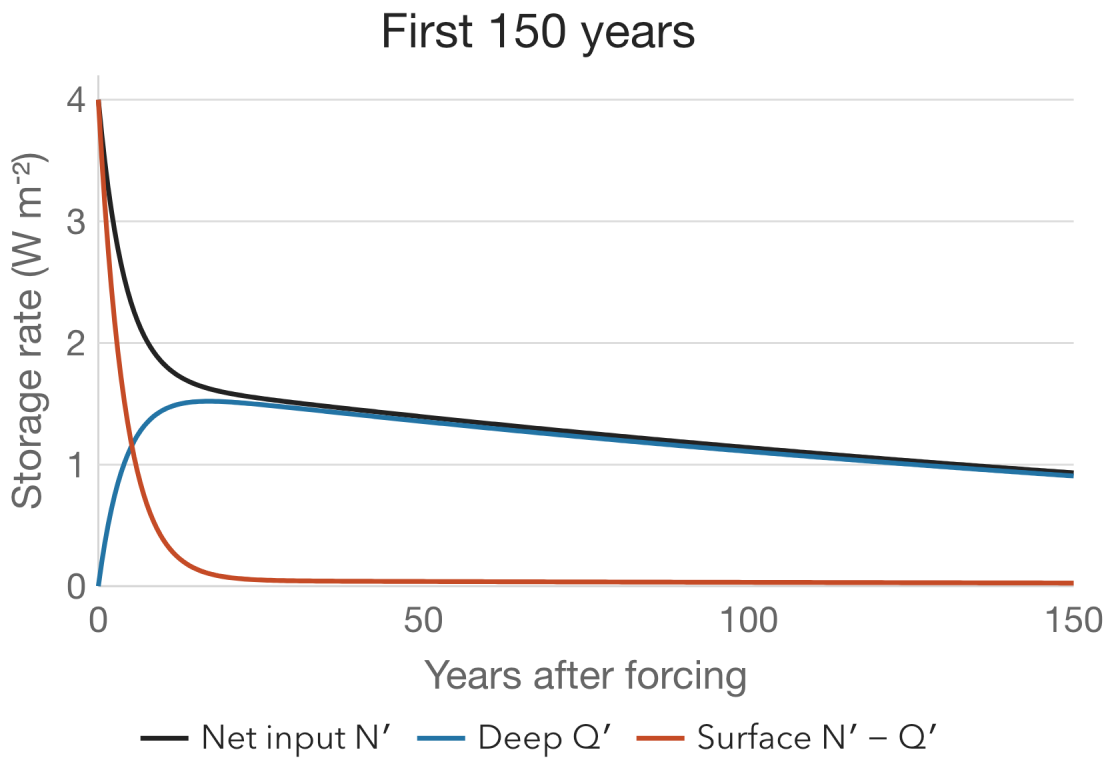


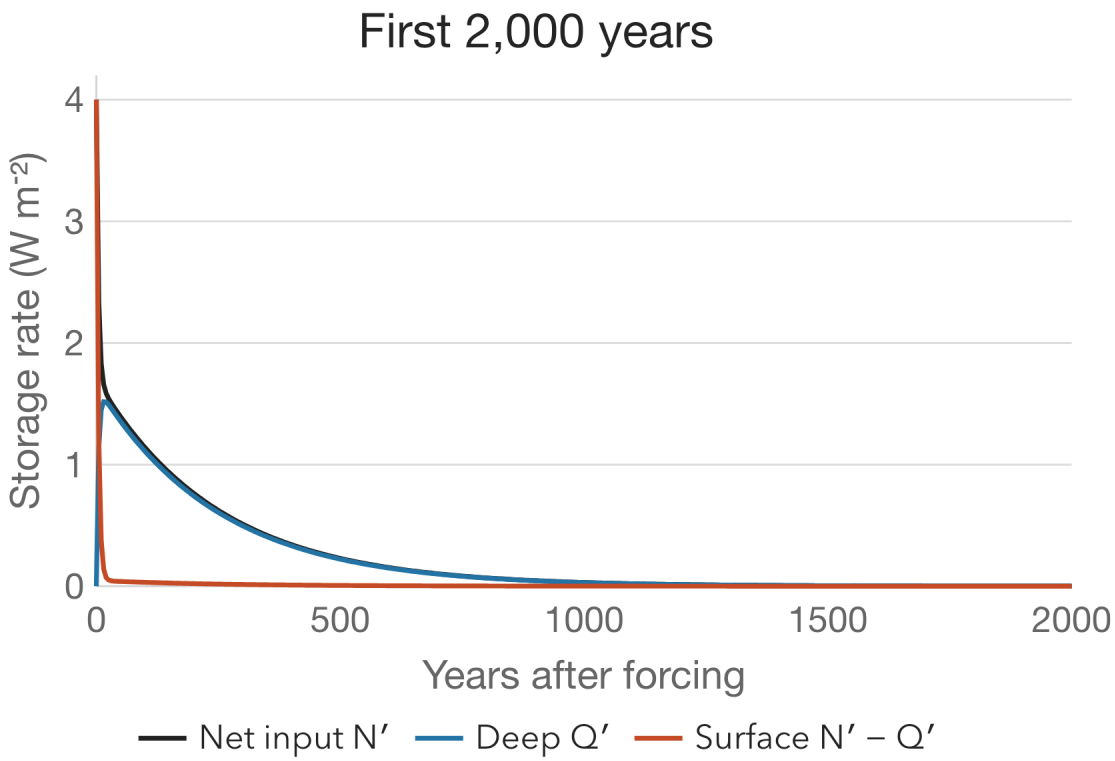


Coupling produces a fast response and a slow tail in surface warming.

$F = 4$ W m^{-2}, $\alpha = -1$ W m^{-2} K^{-1}, $\gamma = 0.7$ W m^{-2} K^{-1} in the coupled model.
Heat capacities: surface 2.2×10^{8}, deep 3.2×10^{9} J m^{-2} K^{-1}, per total Earth area.

Heat uptake rates

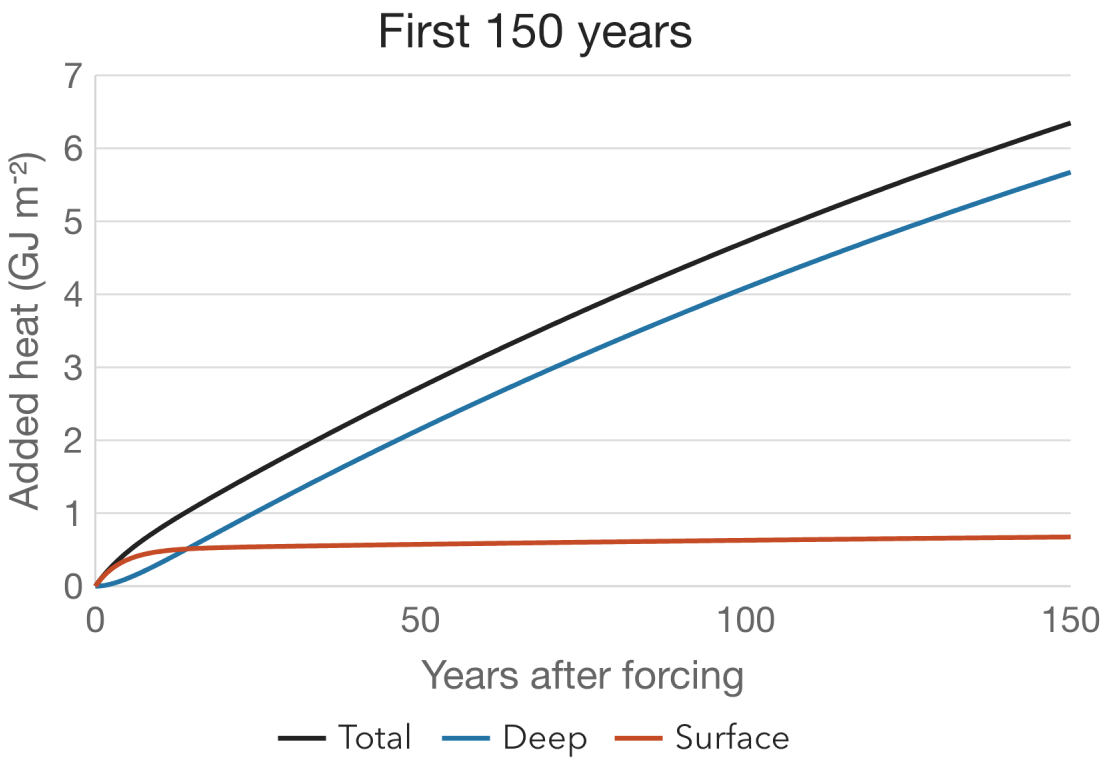


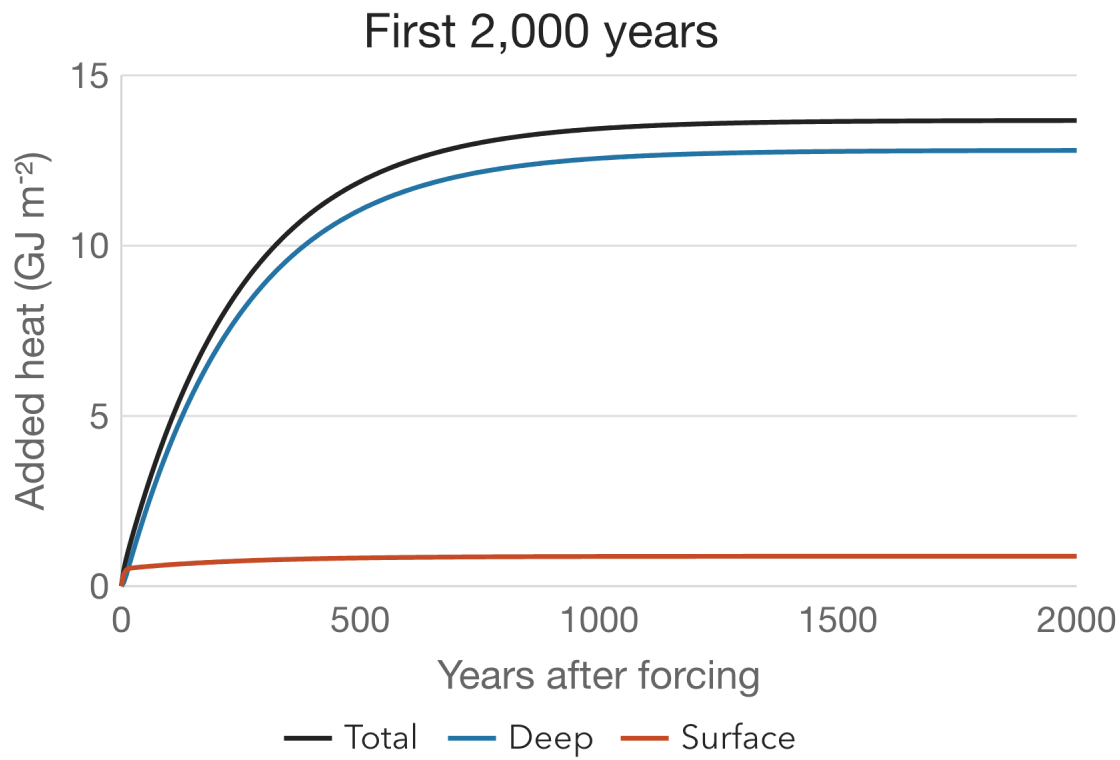


Storage rates: surface N′ – Q′, deep Q′, total N′

Storage rate = dE′/dt = C dT′/dt, in W m^{-2}, for each reservoir.

Accumulated heat





Energy continues to accumulate while the rate of heat uptake declines.

Total stored heat = surface heat + deep-ocean heat

Two-box ocean: equilibrium and adjustment

$$N' = C_s\, dT_s'/dt + C_d\, dT_d'/dt$$

At equilibrium: $T_s' = T_d' = -F/\alpha \quad (\gamma > 0)$

- **Conservation:** internal exchange cancels from the total energy budget.
- **Equilibrium:** vanishing deep storage requires $Q' = 0$. Both warming anomalies eventually match.
- **Adjustment:** exchange and deep heat capacity change the path and pace of warming, while F and α set the endpoint.

Heat capacities and exchange set the adjustment. Forcing and feedback set the endpoint.

Regional climate

A model at every latitude

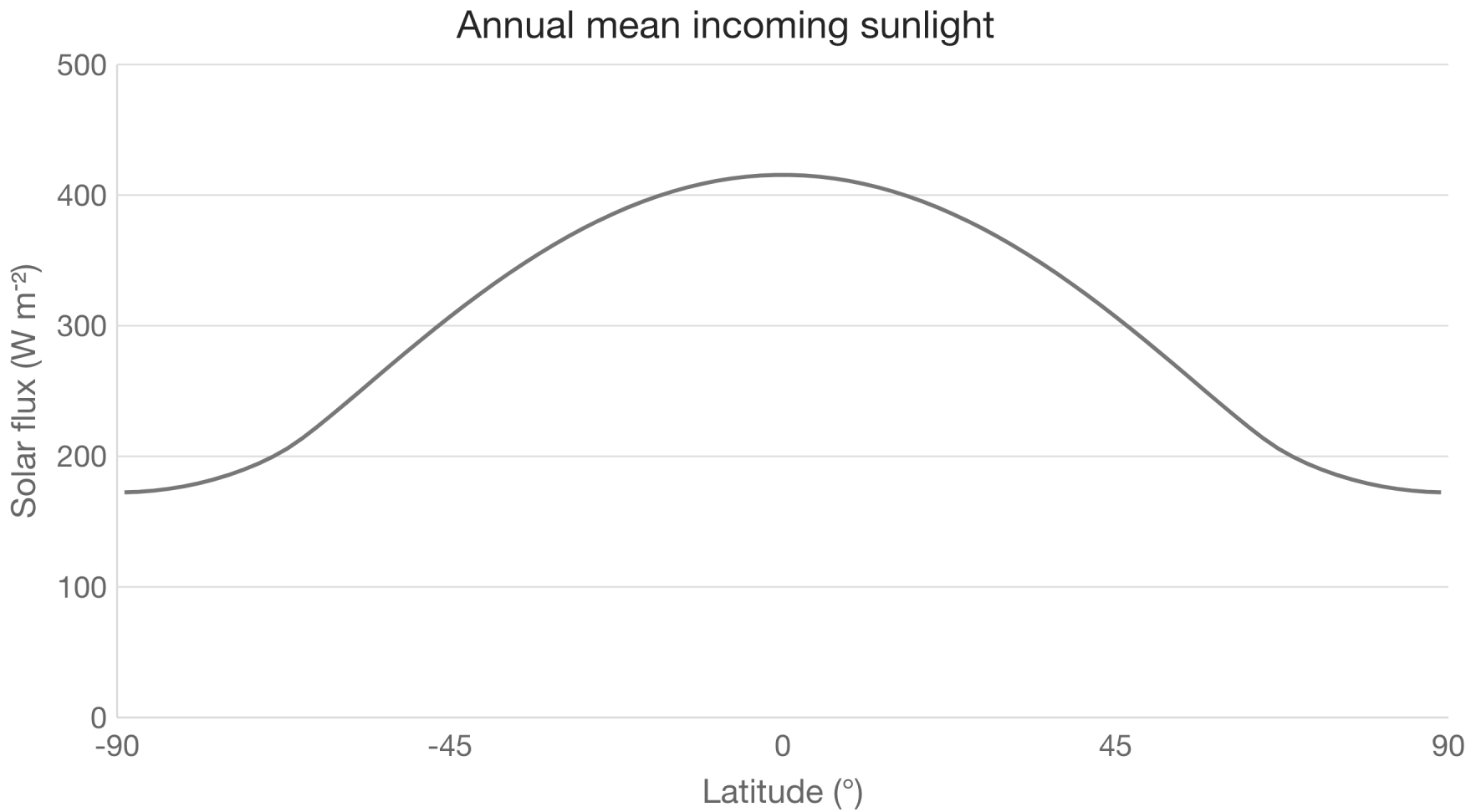
Our earlier EBM matched a global mean temperature using one atmospheric layer.

First apply that same model independently at each latitude, using the local annual mean sunlight.

Does a model fitted to the global mean also reproduce Earth's latitude profile?

First test: constant albedo 0.30, atmospheric emissivity ε = 0.779, and no horizontal transport.

Incoming sunlight across latitude



Our earlier EBM at each latitude

$$ASR(\phi) = (1 - a_0)\, Q(\phi)$$

$$OLR(\phi) = (1 - \varepsilon/2)\, \sigma T_s^4$$

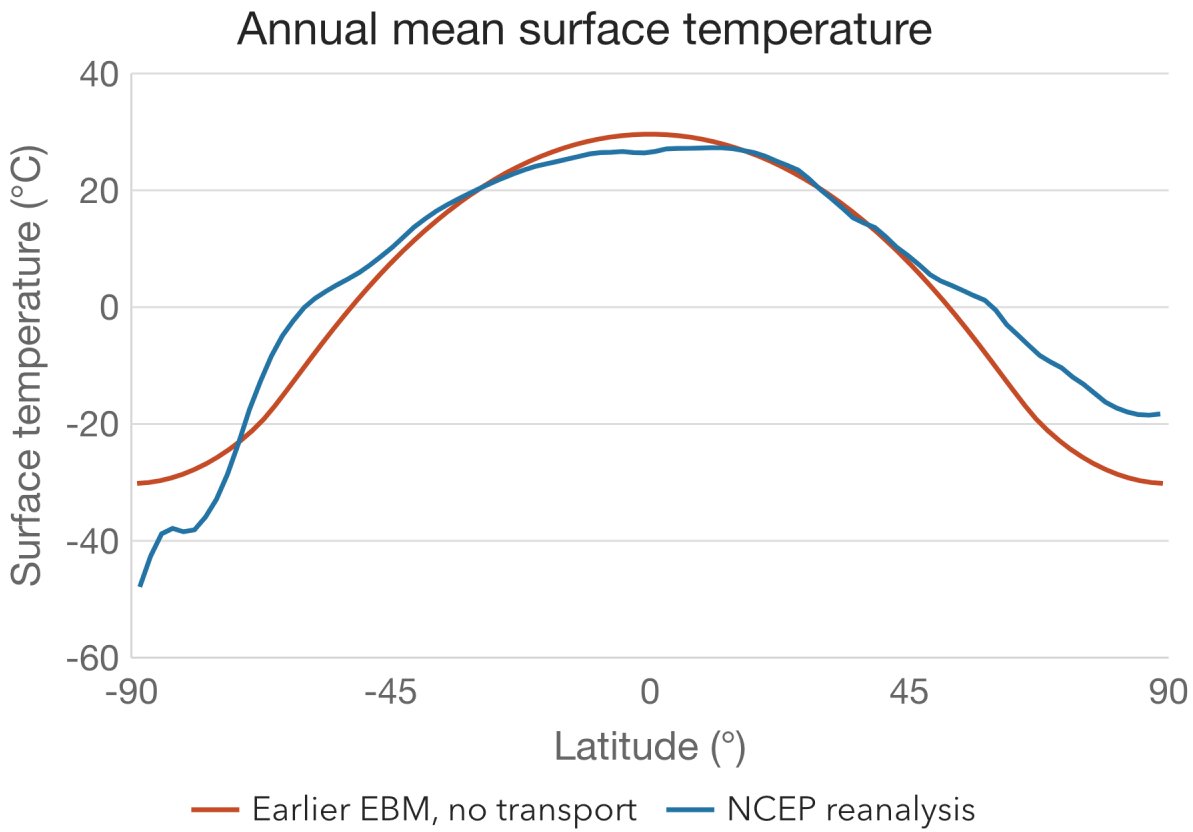
$a_0 = 0.30$ and $\varepsilon = 0.779$ at every latitude

Each band adjusts independently:
$C\, dT_s/dt = ASR - OLR$

The earlier greenhouse effect is retained. Horizontal heat transport is absent.

T_s is in K. ε is atmospheric longwave absorptivity/emissivity. The atmosphere adjusts rapidly.

Testing the predicted temperature profile



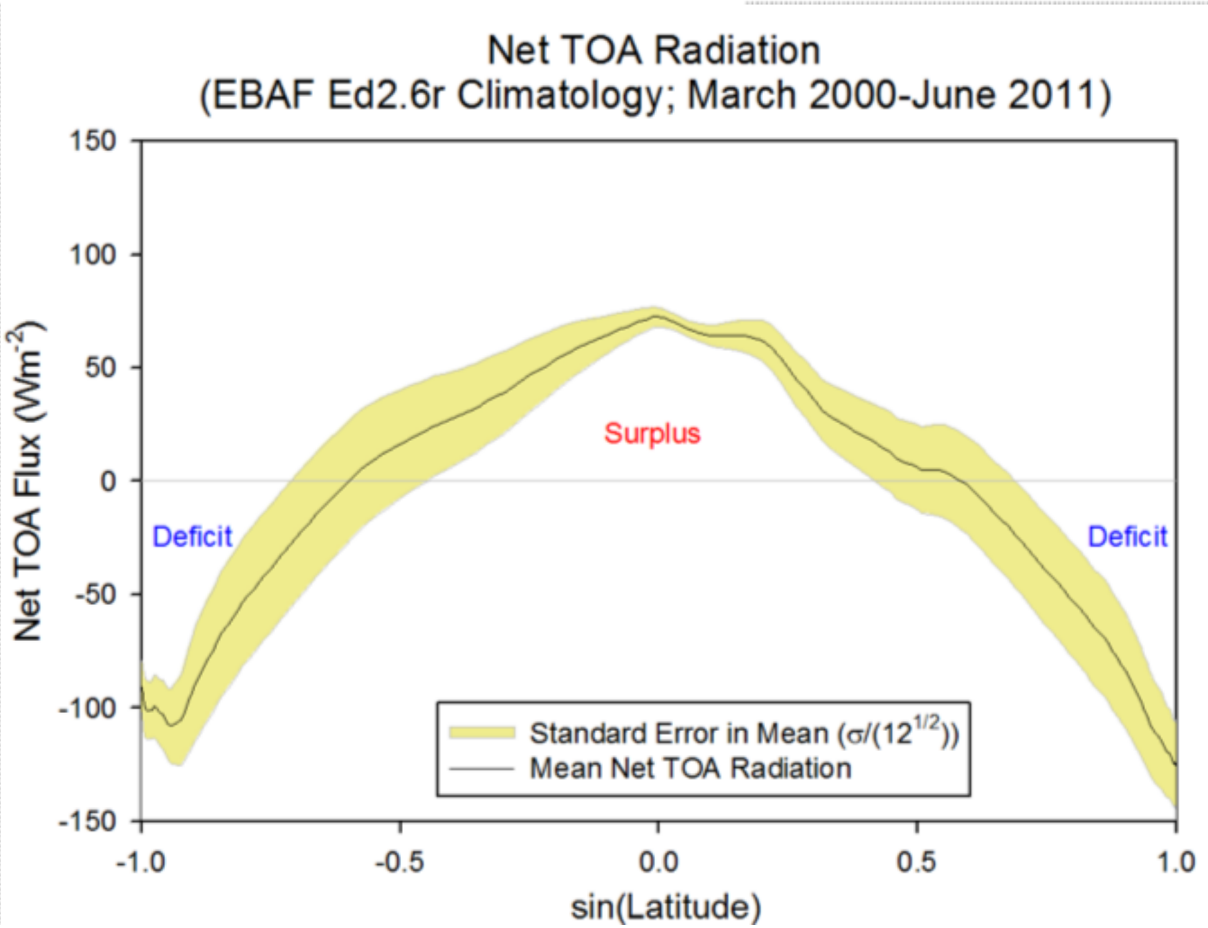
Keep the global calibration:
$a_0 = 0.30$ and $\varepsilon = 0.779$

Start every band at 288 K.
Integrate until local heat storage stops.

Earth's profile has structure and hemispheric differences that this model misses.

Model: 30-year integration. Reanalysis: annual zonal mean surface skin temperature, 1981–2010.

Testing the local radiation budget



sin(latitude): –1 at the South Pole, 0 at the equator, +1 at the North Pole.

Isolated bands at equilibrium:
$N(\phi) = ASR - OLR = 0$

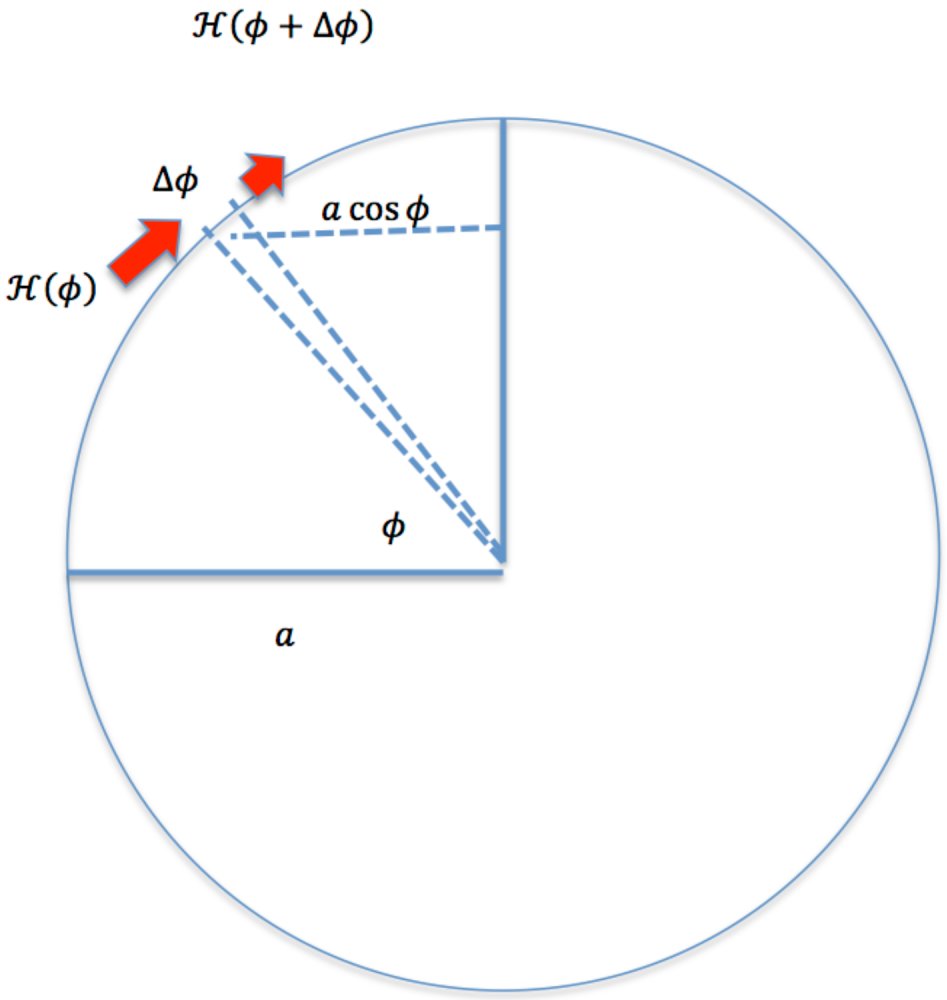
Earth's tropics: positive N
Radiative energy surplus

Higher latitudes: negative N
Radiative energy deficit

What carries surplus energy toward regions with a deficit?

Source: NASA CERES, March 2000–June 2011 climatology.

Energy budget for a latitude band



Each band contains atmosphere and surface together.

T(ϕ,t): surface temperature averaged around a latitude circle.
ϕ denotes latitude.

$$C\,dT/dt = ASR - OLR + h$$

ASR: absorbed sunlight. OLR: outgoing infrared.
h: heating from transport convergence.
ASR, OLR and h use W m^{-2}.
C: heat capacity per area (J m^{-2} K^{-1}).

$$h = [H(\phi) - H(\phi + \Delta\phi)] / A$$

H: northward heat transport, in W (often PW).
A: surface area of the band, in m^2.

Transport convergence and local heating

h = (heat transport in − heat transport out) / band area

Illustrative band area: 5×10^{13} m². 1 PW = 10^{15} W.

Example	Transport in (PW)	Transport out (PW)	Local heating h (W m^{-2})
Equal inflow and outflow	1.0	1.0	0
More heat leaves	1.0	1.5	−10
More heat enters	1.5	1.0	+10

Equal inflow and outflow give $h = 0$, even when transport is large.

h is the local heating rate from transport convergence (W m^{-2}). Values are illustrative.

Regional equilibrium with continuing transport

At steady temperature: $0 = ASR - OLR + h$

Tropical radiative surplus: ASR – OLR = +10 W m^{-2}
Heating from convergence h = –10 W m^{-2} (net export)

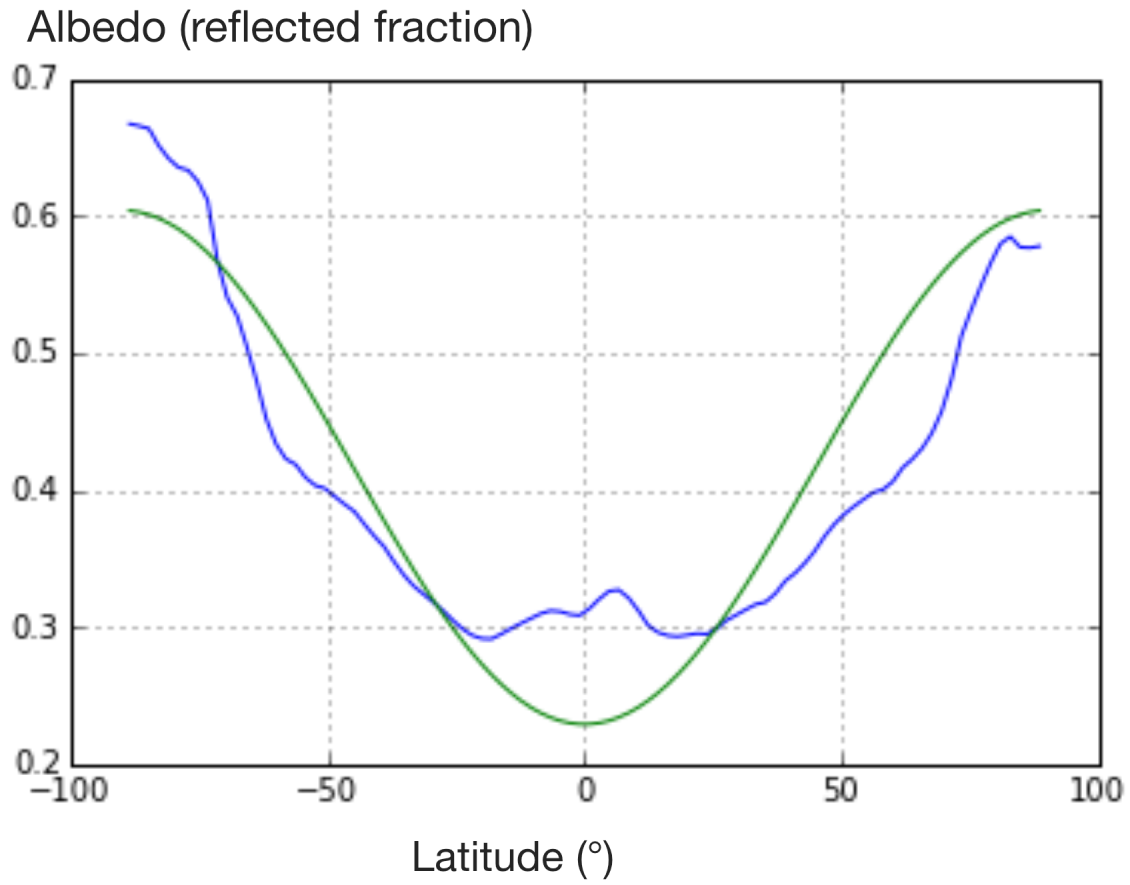
High-latitude radiative deficit: ASR – OLR = –10 W m^{-2}
Heating from convergence h = +10 W m^{-2} (net import)

Steady regional temperatures can coexist with continuing poleward heat transport.

Internal transport cancels from the global energy budget. Local values here are illustrative.

Planetary albedo across latitude

Next refinement: let prescribed albedo vary with latitude.



Blue: NCEP reanalysis

Green: prescribed profile

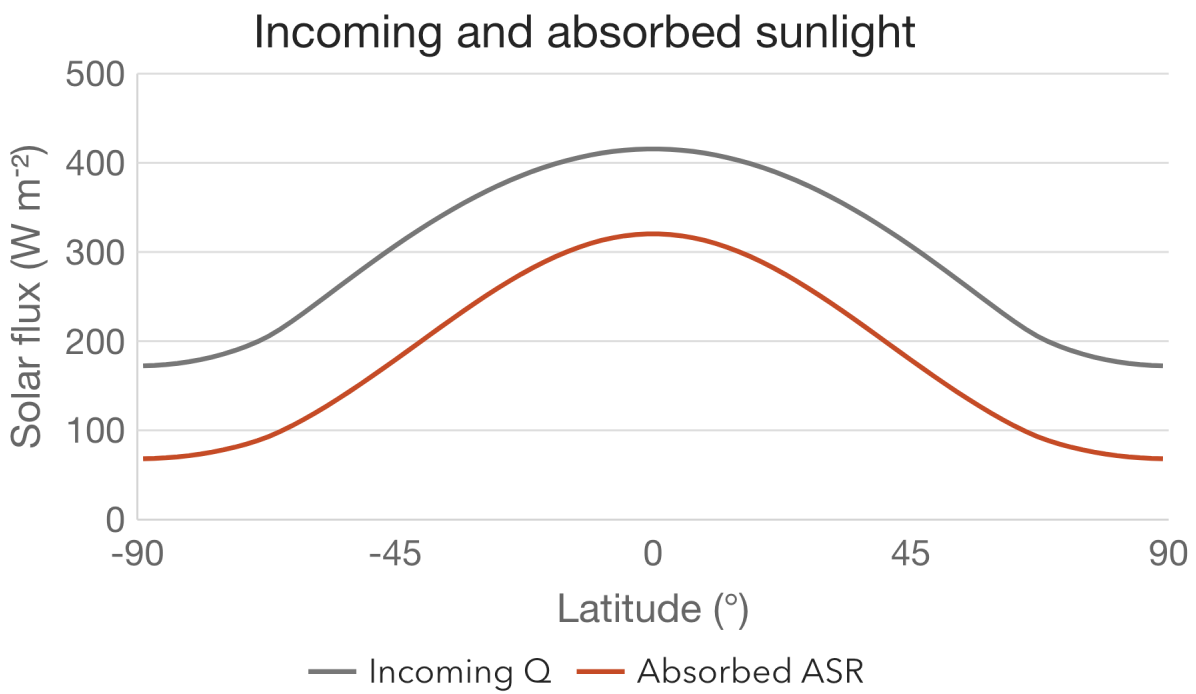
Infer the reflected fraction from TOA incoming and absorbed sunlight: albedo = 1 – ASR / Q

Snow and ice, clouds, and solar angle produce variations missed by constant albedo.

Our smooth, fixed approximation: $\text{albedo}(\phi) = 0.229 + 0.375 \sin^2\phi$

Absorbed sunlight at each latitude

$$ASR(\phi) = [1 - albedo(\phi)]\, Q(\phi)$$



Example: Q = 400 W m^{-2} and albedo = 0.25

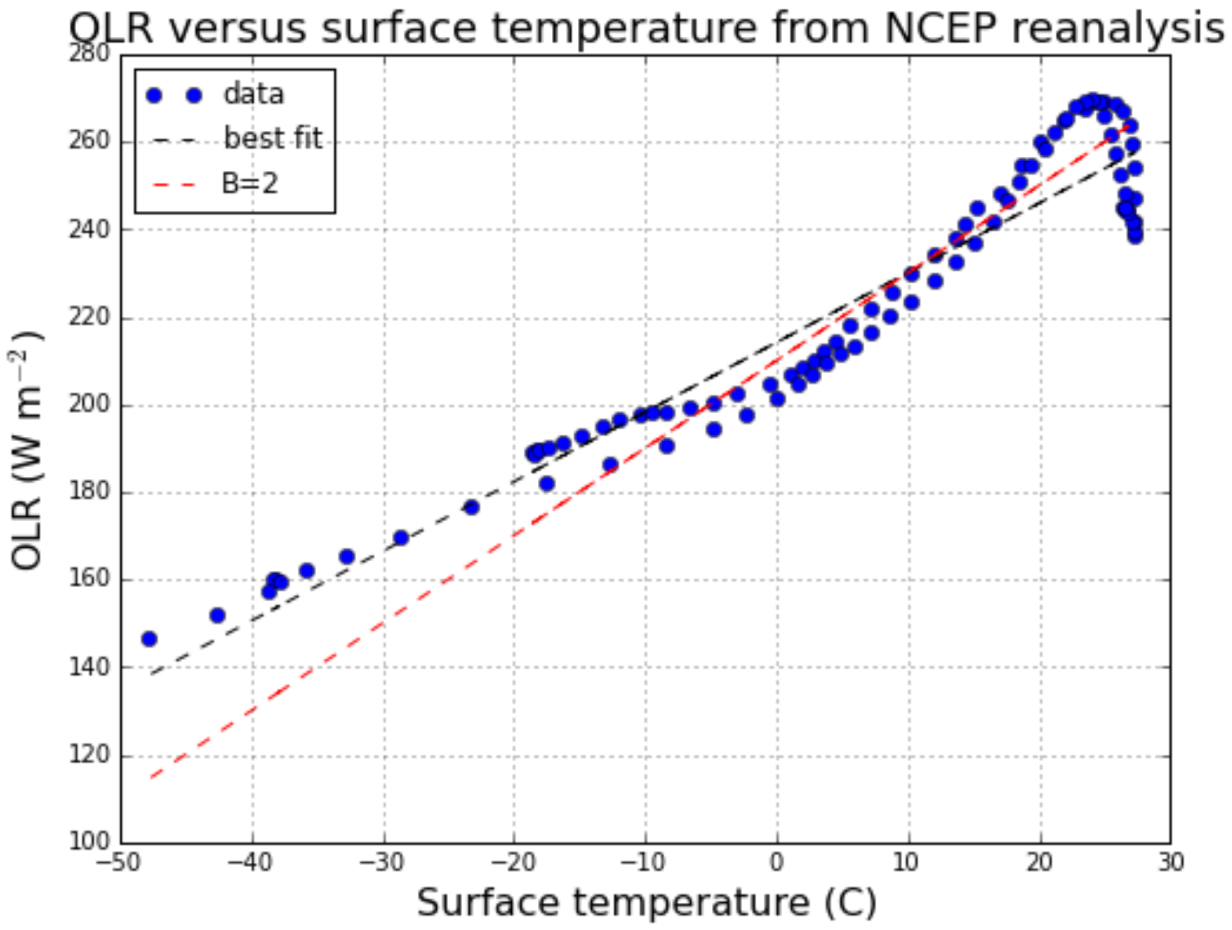
Reflected: 0.25 × 400 = 100 W m^{-2}

Absorbed: 0.75 × 400 = 300 W m^{-2}

ASR is the incoming sunlight that remains after reflection.

This refinement uses the prescribed latitude-dependent albedo. Sunlight remains fixed, with S_0 = 1361 W m^{-2}.

Outgoing infrared radiation



NCEP reanalysis: annual and zonal means.

$$OLR = A + B\,T$$

T is surface temperature in °C.

$A = 210$ W m^{-2}
$B = 2$ W m^{-2} K^{-1}

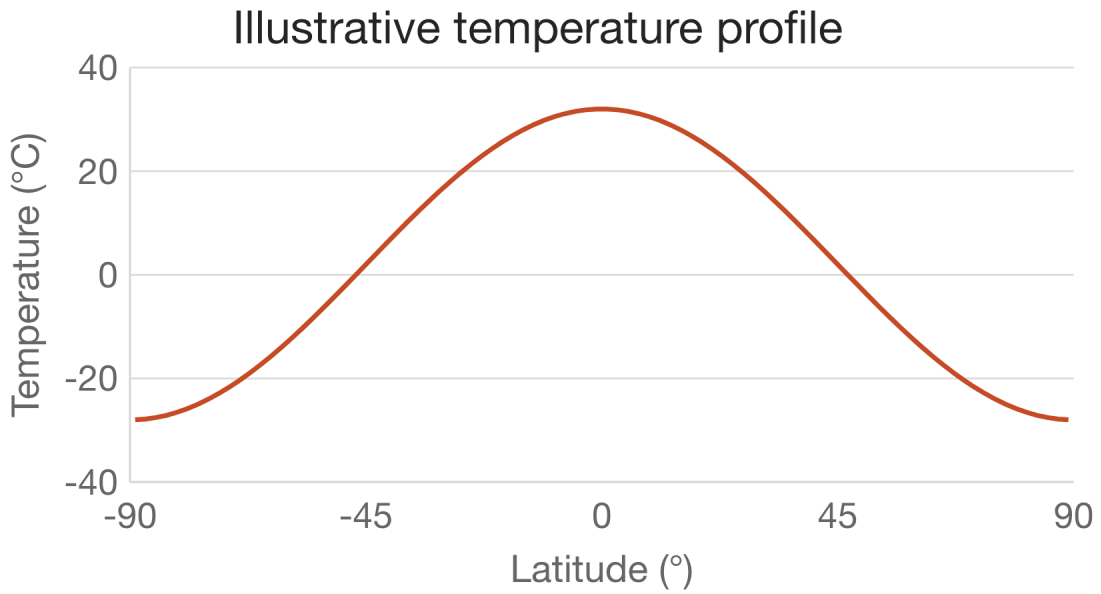
The earlier gray law already includes atmospheric effects.
A + B T provides a simple fit to the regional OLR–surface-T relation.

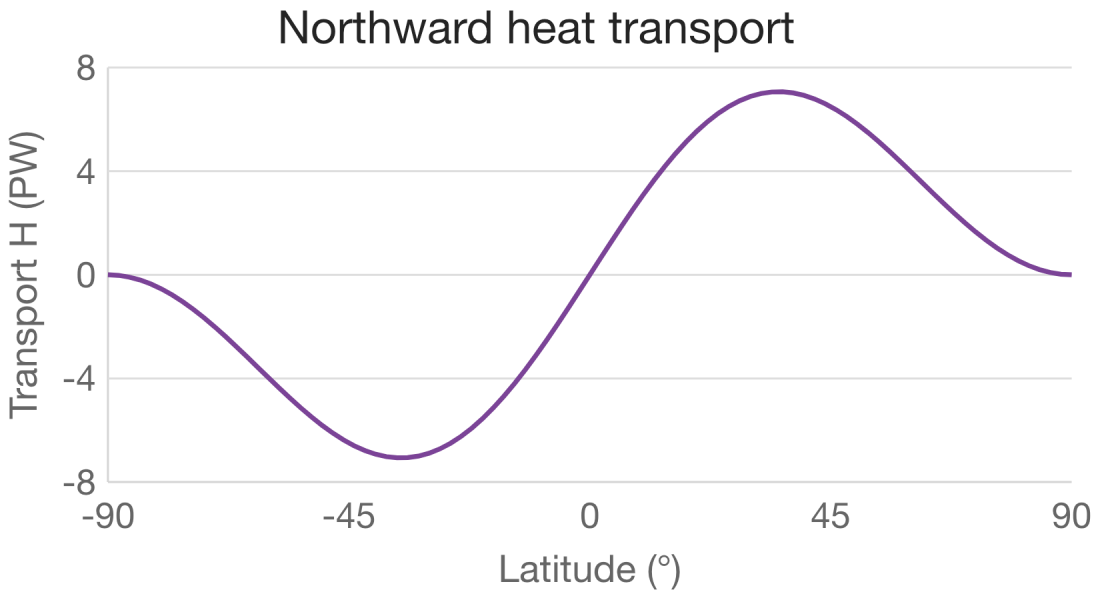
Warming increases OLR and reduces net radiative input.

Heat transport down the temperature gradient

$$H(\phi) = -2\pi R^2 D \cos\phi \, (dT/d\phi)$$

H: northward heat transport (W). D: transport strength (W m^{-2} K^{-1}).
R: Earth radius. Latitude ϕ in radians.



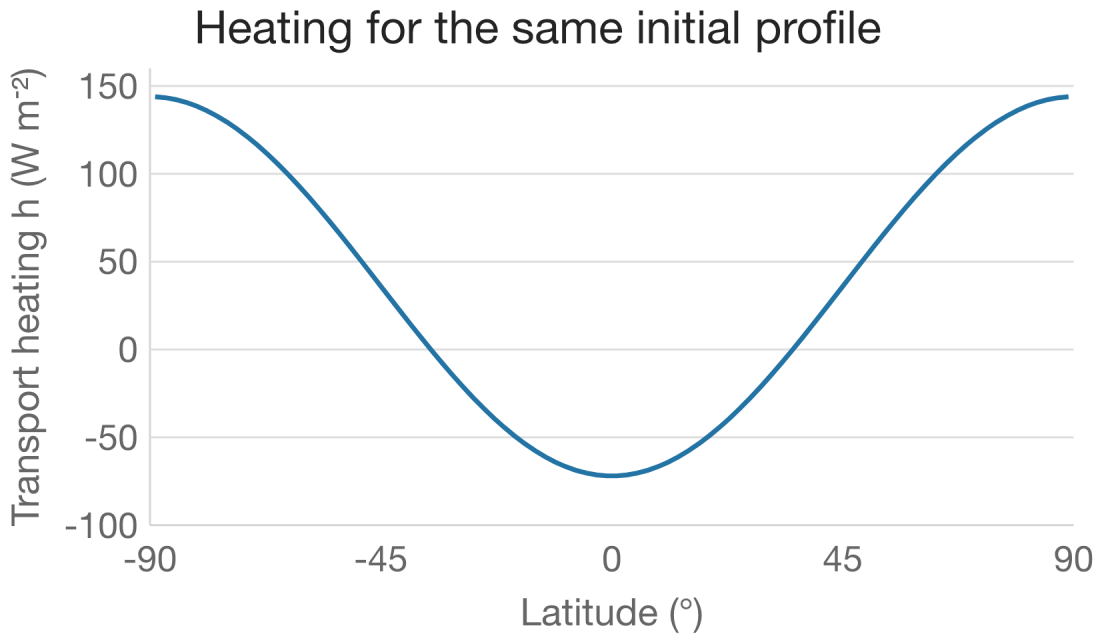


Heat flows toward colder bands. H changes sign across the equator.

Illustrative profile: $T = 32 - 60 \sin^2\phi$ °C. Transport strength D = 0.6 W m^{-2} K^{-1}.

Heating from transport convergence

$h_i = (H_s - H_n)$ / band area



Positive h: net heat gain
Negative h: net heat loss

No transport through the poles.

Heat exchanged between bands cancels in the global budget.

$$C\, dT_i/dt = ASR_i - (A + B\, T_i) + h_i$$

H_s and H_n: northward heat transport at south and north edges (W). Area: m^2. h: W m^{-2}.